

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q12-0128

Northwest Tap 100 kV
August 28, 2013

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q12-0128. The Interconnection Customer proposes a 1500.0 MW natural gas combined cycle facility interconnecting at the Northwest Tap 100 kV substation in the Indiana area. The request was received on May 11, 2012. The study evaluated system intact and single contingency conditions using the regional planning case, including 101 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 19 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$66,827,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

The Barrigada to Northwest Tap 100 KV is overloaded to 800.4% of its 122 MVA emergency rating for the loss of the Northwest Tap to Concord 100 KV L. The project contributes 64.9% DFAX (974.0 MW) to this overload.

Northwest Tap to Concord 100 KV Line: 657.7% loading (145 MVA) following the outage of the Barrigada to Northwest Tap 100 KV Line; DFAX = 66.3%.

The Parkrose to Northwest Tap 100 KV is overloaded to 420.1% of its 124 MVA emergency rating for the loss of the Northwest Tap to Barrigada Substation 100 KV. The project contributes 33.7% DFAX (505.3 MW) to this overload.

Parkrose to Concord Substation 100 KV line: 225.5% loading (135 MVA) following the outage of the Barrigada to Northwest Tap 100 KV Line; DFAX = 22.1%.

The Barrigada to Elkhart Hydro 100 KV is overloaded to 162.9% of its 153 MVA emergency rating for the loss of the Barrigada to Elkhart Hydro 100 KV #2. The project contributes 7.7% DFAX (115.1 MW) to this overload.

For the loss of Barrigada to Elkhart Hydro 100 KV, the Barrigada to Elkhart Hydro 100 KV #2 loads to 162.9% of its 153 MVA rating. Project DFAX: 7.7%.

For the loss of Elkhart Hydro to Lusher 100 KV, the Lusher to Elkhart Hydro 100 KV line loads to 148.2% of its 155 MVA rating. Project DFAX: 7.0%.

The Elkhart Hydro to Lusher 100 KV Circuit 2 is overloaded to 148.2% of its 155 MVA emergency rating for the loss of the Elkhart Hydro to Lusher 100 KV. The project contributes 7.0% DFAX (105.4 MW) to this overload.

The Barrigada to Concord 100 KV Circuit 2 is overloaded to 141.1% of its 165 MVA emergency rating for the loss of the Concord to Northwest Tap 100 KV. The project contributes 23.2% DFAX (348.4 MW) to this overload.

The Barrigada to Concord 100 KV Cir. 1 is overloaded to 134.3% of its 173 MVA emergency rating for the loss of the Northwest Tap to Concord Substation 100 KV. The project contributes 23.2% DFAX (348.4 MW) to this overload.

For the loss of Northwest Tap to Barrigada 100 KV, the Twin Branch Pv to Parkrose 100 KV loads to 129.0% of its 129 MVA rating. Project DFAX: 11.6%.

For the loss of Northwest Tap to Concord 100 KV, the Barrigada to Santa Fe Energy 100 KV loads to 126.4% of its 127 MVA rating. Project DFAX: 5.5%.

Scenic Woods to High Falls - Croghan Ny 100 KV line: 123.7% loading (100 MVA) following the outage of the Barrigada to Northwest Tap 100 KV; DFAX = 11.6%.

The Scenic Woods to Twin Branch Pv 100 KV L is overloaded to 123.6% of its 119 MVA emergency rating for the loss of the Barigada to Northwest Tap 100 KV. The project contributes 11.6% DFAX (174.2 MW) to this overload.

For the loss of Northwest Tap to Concord 100 KV, the C.H. Dexter to Santa Fe Energy 100 KV L loads to 118.8% of its 110 MVA rating. Project DFAX: 5.5%.

The Flood Street to University 230 KV line is overloaded to 113.8% of its 909 MVA emergency rating for the loss of the Flood Street to Cook 500 KV. The project contributes 15.6% DFAX (234.7 MW) to this overload.

Table Mt to Teeland 100 KV L: 113.5% loading (156 MVA) following the outage of the Teeland to Table Mt 100 KV #2; DFAX = 7.0%.

Teeland to Table Mt 100 KV Line 2: 113.5% loading (156 MVA) following the outage of the Teeland to Table Mt 100 KV; DFAX = 7.0%.

For the loss of Lusher to Bear Garden 100 KV, the Sneath-Pacifica Tap to Lusher 100 KV loads to 109.9% of its 154 MVA rating. Project DFAX: 6.1%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Northwest Tap to Barrigada 100 KV	Rebuild line with higher-capacity conductor	\$1,411,000	95% (\$1,346,000)
Northwest Tap to Concord 100 KV Line	Rebuild line with higher-capacity conductor	\$2,884,000	96% (\$2,754,000)
Parkrose to Northwest Tap 100 KV	Rebuild line with higher-capacity conductor	\$2,372,000	92% (\$2,170,000)
Concord to Parkrose 100 KV	Rebuild line with higher-capacity conductor	\$7,381,000	89% (\$6,588,000)
Elkhart Hydro to Barrigada Substation 100 KV Ckt 1	Rebuild line with higher-capacity conductor	\$727,000	100% (\$727,000)
Barrigada to Elkhart Hydro 100 KV Line 2	Rebuild line with higher-capacity conductor	\$811,000	100% (\$811,000)
Lusher to Elkhart Hydro 100 KV Line 1	Rebuild line with higher-capacity conductor	\$3,067,000	100% (\$3,067,000)
Elkhart Hydro to Lusher 100 KV Circuit 2	Rebuild line with higher-capacity conductor	\$2,339,000	100% (\$2,339,000)
Barrigada to Concord 100 KV #2	Rebuild line with higher-capacity conductor	\$3,890,000	100% (\$3,890,000)
Barrigada to Concord 100 KV Cir. 1	Rebuild line with higher-capacity conductor	\$3,317,000	100% (\$3,317,000)
Twin Branch Pv to Parkrose 100 KV	Rebuild line with higher-capacity conductor	\$5,434,000	67% (\$3,625,000)
Barrigada to Santa Fe Energy 100 KV	Rebuild line with higher-capacity conductor	\$4,058,000	100% (\$4,058,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Scenic Woods to High Falls - Croghan Ny 100 KV	Rebuild line with higher-capacity conductor	\$2,900,000	67% (\$1,935,000)
Scenic Woods Substation to Twin Branch Pv Substation 100 KV L	Rebuild line with higher-capacity conductor	\$1,997,000	67% (\$1,332,000)
C.H. Dexter to Santa Fe Energy 100 KV L	Rebuild line with higher-capacity conductor	\$12,224,000	100% (\$12,224,000)
Flood Street to University 230 KV line	Rebuild line with higher-capacity conductor	\$31,000,000	48% (\$14,797,000)
Teeland to Table Mt 100 KV Cir. 1	Rebuild line with higher-capacity conductor	\$1,424,000	38% (\$538,000)
Table Mt to Teeland 100 KV line	Rebuild line with higher-capacity conductor	\$691,000	38% (\$261,000)
Sneath-Pacifica Tap to Lusher 100 KV	Reconductor line	\$1,048,000	100% (\$1,048,000)
	Total allocated to project		\$66,827,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.